IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program

WEEK 02 ASSIGNMENT REPORT

Computer Networking Fundamentals & Network Configuration

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1. Introduction

Computer networks form the backbone of nearly every modern system, connecting devices so they can share data, services, and resources. For a Cyber Security Analyst, understanding how networks are configured and how traffic moves between devices is a prerequisite to detecting anomalies, investigating incidents, and hardening infrastructure.

This report documents the practical and theoretical work completed for Week 2 of the IT-Simplera Cyber Security Analyst Internship Program. It covers how to read a device's network configuration, the purpose of five essential command-line diagnostic tools, a comparison of routers and switches, a comparison of IPv4 and IPv6 addressing, and a breakdown of the seven-layer OSI Model.

2. Task 1: Network Information (Q1)

The ipconfig /all command displays the full TCP/IP configuration for every network adapter on a Windows machine. Running it is the fastest way to confirm how a device is currently identified and addressed on its network. The three values below are the ones most relevant to day-to-day troubleshooting and security analysis.

IPv4 Address

The IPv4 address is the logical, 32-bit address assigned to the device on its local network (e.g., 192.168.1.10). It identifies the device so that other devices and the router know where to deliver traffic. It is used for routing decisions at Layer 3 of the OSI Model.

Physical (MAC) Address

The MAC address is a 48-bit hardware address burned into the network interface card by its manufacturer (e.g., 00-1A-2B-3C-4D-5E). It uniquely identifies the physical adapter and is used for delivering frames within the local network segment at Layer 2 of the OSI Model.

Default Gateway

The default gateway is the IP address of the router that connects the local network to other networks, including the internet (e.g., 192.168.1.1). Any traffic destined for an address outside the local subnet is forwarded to this gateway first.

```
C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Bunny
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 74-5D-22-60-D7-63
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 74-13-EA-3A-48-38
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 76-13-EA-3A-48-37
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::247f:4d94:b75f:342e%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.247.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, 9 July 2026 01:13:40
   Lease Expires . . . . . . . . . . : Thursday, 9 July 2026 01:43:39
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.247.254
   DHCPv6 IAID . . . . . . . . . . . : 687886422
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-5C-2E-12-74-5D-22-60-D7-63
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::c47f:62f5:e014:1575%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.91.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, 9 July 2026 01:13:41
   Lease Expires . . . . . . . . . . : Thursday, 9 July 2026 01:58:40
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.91.254
   DHCPv6 IAID . . . . . . . . . . . : 704663638
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-5C-2E-12-74-5D-22-60-D7-63
   Primary WINS Server . . . . . . . : 192.168.91.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
   Physical Address. . . . . . . . . : 74-13-EA-3A-48-37
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::f9b9:6b09:bbed:c657%10(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.100.203(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Wednesday, 8 July 2026 19:00:40
   Lease Expires . . . . . . . . . . : Friday, 10 July 2026 01:13:41
   Default Gateway . . . . . . . . . : 192.168.100.1
   DHCP Server . . . . . . . . . . . : 192.168.100.1
   DHCPv6 IAID . . . . . . . . . . . : 125047786
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-5C-2E-12-74-5D-22-60-D7-63
   DNS Servers . . . . . . . . . . . : 192.168.100.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

C:\Windows\System32>
```

3. Task 2: Basic Networking Commands (Q2)

Each of the five commands below was executed from the command prompt. The table summarizes the purpose of each; a screenshot placeholder follows for the corresponding output.

Command	Purpose
ipconfig /all	Displays complete IP configuration details (IPv4/IPv6 address, subnet mask, default gateway, MAC address, DNS servers) for every adapter on the machine.
ping google.com	Tests reachability of a remote host by sending ICMP echo requests and measuring round-trip time, confirming basic connectivity.
tracert google.com	Traces the path packets take to a destination, listing every intermediate router (hop) and the latency at each one, useful for locating where a connection is failing or slow.
nslookup google.com	Queries a DNS server to resolve a domain name into its corresponding IP address, helping verify that DNS resolution is working correctly.
arp -a	Displays the ARP cache, which maps local IP addresses to their corresponding MAC addresses, showing which devices the computer has recently communicated with on the LAN.

```
C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Bunny
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 74-5D-22-60-D7-63
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 74-13-EA-3A-48-38
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 76-13-EA-3A-48-37
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::247f:4d94:b75f:342e%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.247.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Thursday, 9 July 2026 01:13:40
   Lease Expires . . . . . . . . . . : Thursday, 9 July 2026 01:43:39
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.247.254
   DHCPv6 IAID . . . . . . . . . . . : 687886422
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-5C-2E-12-74-5D-22-60-D7-63
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

```
C:\Windows\System32>ping google.com

Pinging google.com [142.250.200.174] with 32 bytes of data:
Reply from 142.250.200.174: bytes=32 time=46ms TTL=116
Reply from 142.250.200.174: bytes=32 time=47ms TTL=116
Reply from 142.250.200.174: bytes=32 time=46ms TTL=116
Reply from 142.250.200.174: bytes=32 time=48ms TTL=116

Ping statistics for 142.250.200.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 46ms, Maximum = 48ms, Average = 46ms

C:\Windows\System32>
```

```
C:\Windows\System32>tracert google.com

Tracing route to google.com [142.250.200.174]
over a maximum of 30 hops:

  1     9 ms     3 ms     3 ms  192.168.100.1
  2     2 ms     1 ms     4 ms  202.70.146.130
  3     6 ms     4 ms     4 ms  10.253.13.50
  4    46 ms    44 ms    45 ms  74.125.118.170
  5    47 ms    47 ms    46 ms  172.253.51.205
  6    47 ms    45 ms    47 ms  74.125.37.203
  7    52 ms    47 ms    50 ms  pnfjra-aq-in-f14.1e100.net [142.250.200.174]

Trace complete.
```

```
C:\Windows\System32>nslookup google.com
Server:  UnKnown
Address:  192.168.100.1

Non-authoritative answer:
Name:    google.com
Addresses:  2a00:1450:4019:80f::200e
          142.250.200.174


C:\Windows\System32>
```

```
C:\Windows\System32>arp -a

Interface: 192.168.91.1 --- 0x2
  Internet Address      Physical Address      Type
  192.168.91.254        00-50-56-ea-a5-a9     dynamic
  192.168.91.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.100.203 --- 0xa
  Internet Address      Physical Address      Type
  192.168.100.1         14-8c-4a-fc-42-4a     dynamic
  192.168.100.13        d8-cb-8a-93-93-78     dynamic
  192.168.100.23        98-90-96-bb-b1-9f     dynamic
  192.168.100.43        18-66-da-31-79-cc     dynamic
  192.168.100.93        98-90-96-b7-07-2e     dynamic
  192.168.100.118       6c-3b-e5-38-c2-ff     dynamic
  192.168.100.184       b0-7b-25-56-28-fe     dynamic
  192.168.100.189       34-73-5a-b2-ad-0b     dynamic
  192.168.100.205       68-f7-28-19-e8-91     dynamic
  192.168.100.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.247.1 --- 0x12
  Internet Address      Physical Address      Type
  192.168.247.254       00-50-56-f0-a5-ec     dynamic
  192.168.247.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Windows\System32>
```

4. Task 3: Router vs. Switch (Q3)

Routers and switches both forward traffic, but they operate at different layers of the OSI Model and serve different purposes in a network, as summarized below.

Comparison Point	Router	Switch
Primary Function	Directs data between different networks (e.g., LAN to internet)	Connects multiple devices within the same local network
OSI Layer	Network Layer (Layer 3)	Data Link Layer (Layer 2)
Addressing Used	IP address	MAC address
Data Unit	Packet	Frame
Broadcast Domains	Separates broadcast domains between networks	Devices share a single broadcast domain (unless VLANs are configured)
Typical Port Count	Few ports (commonly 2-5: WAN + LAN)	Many ports (commonly 4-48+)
Common Use Case	Connecting a home or office LAN to the internet, or linking multiple networks	Connecting PCs, printers, and servers within one LAN

5. Task 4: IPv4 vs. IPv6 (Q4)

IPv6 was developed primarily to address the exhaustion of IPv4 addresses and to improve on its design. Key differences are summarized below.

Feature	IPv4	IPv6
Address Length	32-bit	128-bit
Address Format	Dotted-decimal (four numbers separated by dots)	Hexadecimal, separated by colons
Total Address Space	About 4.3 billion addresses	About 340 undecillion addresses
Header Design	More complex, variable length with optional fields	Simplified, fixed-length header for faster processing
Built-in Security	Not built in; IPSec is optional	IPSec support is built into the protocol
Address Configuration	Manual configuration or DHCP	Supports stateless auto-configuration (SLAAC) or DHCPv6
NAT Requirement	Commonly required due to limited address space	Not typically required, given the vast address space

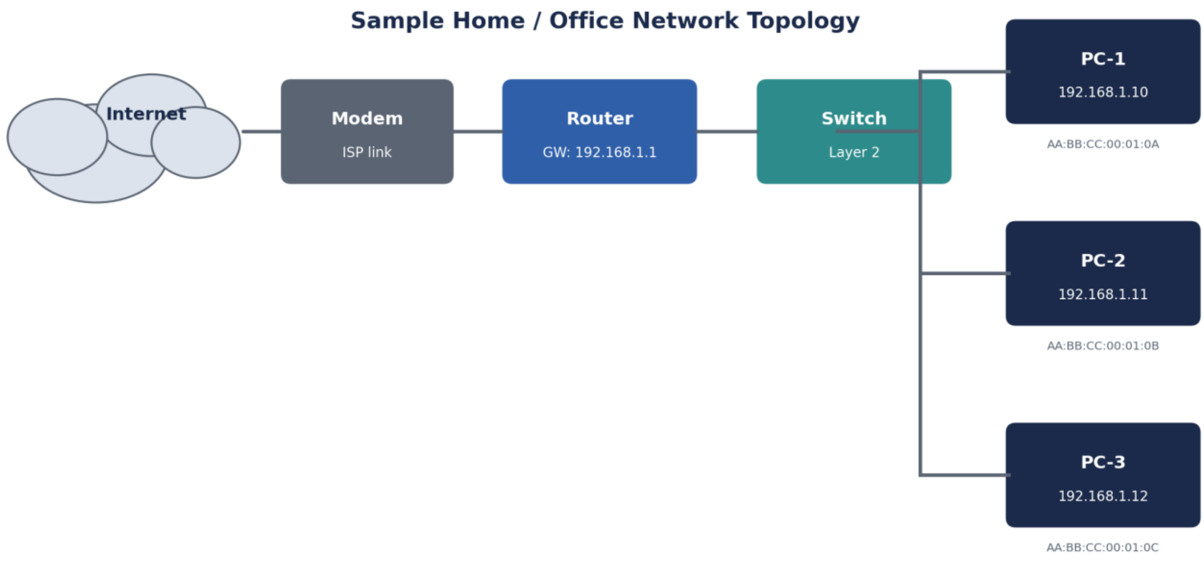
6. Task 5: The OSI Model (Q5)

The OSI Model breaks network communication into seven layers. The layers are listed below in order from Layer 7 (closest to the user) down to Layer 1 (the physical medium).

Layer	Primary Function	Common Protocol	Associated Device
7. Application	Provides network services directly to end-user applications	HTTP / DNS	Gateway
6. Presentation	Translates, encrypts, and compresses data between systems	TLS/SSL	Gateway
5. Session	Establishes, manages, and terminates communication sessions	NetBIOS	Gateway
4. Transport	Provides reliable, ordered (or fast, best-effort) end-to-end delivery	TCP / UDP	Firewall
3. Network	Handles logical addressing and routing between networks	IP	Router
2. Data Link	Handles node-to-node delivery and physical addressing within a segment	Ethernet	Switch / Bridge
1. Physical	Transmits raw bits over a physical medium (cables, radio signals)	Ethernet (physical signaling)	Hub / Repeater

7. Network Diagram

The diagram below illustrates a simple home or office network topology, showing how a router and switch sit between the internet connection and the end-user devices, along with an example private IP addressing scheme.



8. Challenges Faced

- Distinguishing which OSI layer a real device belongs to, since many modern devices (e.g., multi-layer switches, firewalls) operate across more than one layer in practice.
- Interpreting tracert output when some hops show "Request timed out," which usually means an intermediate router or firewall is blocking or deprioritizing ICMP traffic rather than indicating an outage.
- Remembering IPv6 shorthand rules, such as omitting leading zeros and using a double colon once per address to represent consecutive blocks of zeros.
- Understanding how VLANs change a switch's default single-broadcast-domain behavior.

9. Conclusion

This assignment reinforced the foundational building blocks of computer networking that underpin cybersecurity work: reading a device's own configuration, using command-line tools to test and trace connectivity, understanding the distinct roles of routers and switches, comparing IPv4 and IPv6 addressing, and mapping network activity onto the seven layers of the OSI Model. Together, these skills provide the baseline needed to recognize normal network behavior, which is a prerequisite for spotting abnormal or malicious activity in future weeks of the internship.

10. References

- Cisco Networking Academy — Introduction to Networking course materials
- IETF RFC 791 — Internet Protocol (IPv4)
- IETF RFC 8200 — Internet Protocol, Version 6 (IPv6) Specification
- Microsoft Learn Documentation — Windows networking command-line tools (ipconfig, ping, tracert, nslookup, arp)
- ISO/IEC 7498-1 — OSI Reference Model